



DP-600 : Implement analytics solutions using Microsoft Fabric

Exercise 15 : Build an ontology from a semantic model in Fabric IQ

- **Create a workspace**
- **Create a lakehouse with sample data**
- **Create an eventhouse with streaming data**
- **Create a semantic model**
- **Generate the ontology**
- **Review and add entity type keys**
- **Verify and configure relationship types**
- **Add time-series binding to VitalSignEquipment**
- **Preview the ontology**



DP-600 : Implement analytics solutions using Microsoft Fabric

➤ Create a workspace

Trials activated: 10 days left

Create a workspace

Name *

ontology_semantic_model

✓ This name is available

Description

Describe this workspace

Domain ⓘ

Assign to a domain (optional)

[Learn more about workspace settings](#)

Workspace image

 [Upload](#) [Reset](#)

Advanced ^

Contact list * ⓘ

 dagstudentsgreatlearning (Owner) [×](#)

Workspace type ⓘ

What a user can do in a workspace depends on the individual user license(s) assigned to that user and the workspace type. [Learn more](#)

— Power BI only workspace types
Power BI workspaces only support the use of Power BI capabilities.

[Apply](#) [Cancel](#)



DP-600 : Implement analytics solutions using Microsoft Fabric



Create a lakehouse with sample data

New Lakehouse

Name *

LamnaHealthcareLH

Location *

ontology_semantic_model

☒ Lakehouse schemas ⓘ

Create **Cancel**

Explorer

Add lakehouses

LamnaHealthcareLH

- Tables
 - dbo
 - departments
 - hospitals
 - patients
 - rooms
 - vitalsignequipment
 - Files

Files

ontology_semantic_model > LamnaHealthcareLH > Files

Showing 5 items

Name	Date modified	Type	Size
Departments.csv	6/12/2026, 12:46:...	csv	126 B
Hospitals.csv	6/12/2026, 12:46:...	csv	91 B
Patients.csv	6/12/2026, 12:46:...	csv	287 B
Rooms.csv	6/12/2026, 12:46:...	csv	273 B
VitalSignEquipment.csv	6/12/2026, 12:46:...	csv	309 B



DP-600 : Implement analytics solutions using Microsoft Fabric

➤ Create an eventhouse with streaming data

New Eventhouse

Eventhouse name

LamnaHealthcareEH

CreateCancel

Home

Workspaces

Copilot

OneLake catalog

Monitor

Real-Time

Workloads

ontology_se
mantic_m...

My workspace

...

Fabric

Eventhouse Database Table

Explore data Get data Query with code Real-Time Dashboard KQL Queryset Power BI report Anomaly Detector Notebook View files OneLake Rename Edit script

LamnaHealthcareEH

System overview

Databases

Monitoring

Search

KQL databases

LamnaHealthcareEH

LamnaHealthcareEH_queryset

Tables

VitalSignsReadings

Ingestion Mappings

Update Policy

ReadingId

EquipmentId

Timestamp

Database / Tables / VitalSignsReadings

Overview Entity diagram

Data Activity Tracker

Ingestion: 20 rows Last run: 12:58:38 AM

1H 6H 1D 3D 7D 30D Interval: 12 hours

20

0

Jun 08, 12:00 PM Jun 09, 12:00 AM Jun 09, 12:00 PM Jun 10, 12:00 AM Jun 10, 12:00 PM Jun 11, 12:00 AM Jun 11, 12:00 PM

Data preview Schema insights

Search keyword

ReadingId	EquipmentId	Timestamp	HeartRate	OxygenSatura...	RespiratoryR...	IngestionTime
> 4	VS-1002	2026-02-02T08:00:00.000	92	99	14	2026-06-11T19:28:27.65535
> 7	VS-1003	2026-02-02T08:00:00.000	65	94	12	2026-06-11T19:28:27.65535
> 10	VS-1004	2026-02-02T08:00:00.000	88	99	15	2026-06-11T19:28:27.65535
> 13	VS-1005	2026-02-02T08:00:00.000	75	98	14	2026-06-11T19:28:27.65535
> 1	VS-1001	2026-02-02T08:00:00.000	78	98	16	2026-06-11T19:28:27.65535
> 2	VS-1001	2026-02-02T08:05:00.000	82	97	18	2026-06-11T19:28:27.65535
> 5	VS-1002	2026-02-02T08:05:00.000	95	98	15	2026-06-11T19:28:27.65535
> 8	VS-1003	2026-02-02T08:05:00.000	68	95	13	2026-06-11T19:28:27.65535
> 11	VS-1004	2026-02-02T08:05:00.000	90	98	16	2026-06-11T19:28:27.65535

Top 20 records from the last 365 Days, limited by the hot cache retention

Table details



DP-600 : Implement analytics solutions using Microsoft Fabric



Create a semantic model

New semantic model

Direct Lake semantic model name ⓘ

Workspace ⓘ

ontology_semantic_model

Select or deselect tables for the semantic model.

SQL views cannot be selected for Direct Lake on OneLake, but tables based on SQL views can be added to the semantic model later using other storage modes.

☒ Select all

☒ dbo

☒ departments

☒ hospitals

☒ patients

☒ rooms

☒ vitalsignequipment

[Learn more about Direct Lake](#)

Confirm

Cancel



DP-600 : Implement analytics solutions using Microsoft Fabric

➤ Create a semantic model

Manage relationships

+ New relationship

Edit

Delete

Filter

☐ From: table (column) ↑	Relationship	To: table (column)	Status	
☐ departments (HospitalId)	* — ◀▶ — 1	hospitals (HospitalId)	Active	...
☐ rooms (DepartmentId)	* — ◀▶ — 1	departments (DepartmentId)	Active	...
☐ rooms (RoomId)	* — ◀▶ — 1	patients (CurrentRoomId)	Active	...
☐ vitalsignequipment (PatientId)	* — ◀▶ — 1	patients (PatientId)	Active	...



DP-600 : Implement analytics solutions using Microsoft Fabric



Generate the ontology

New Ontology

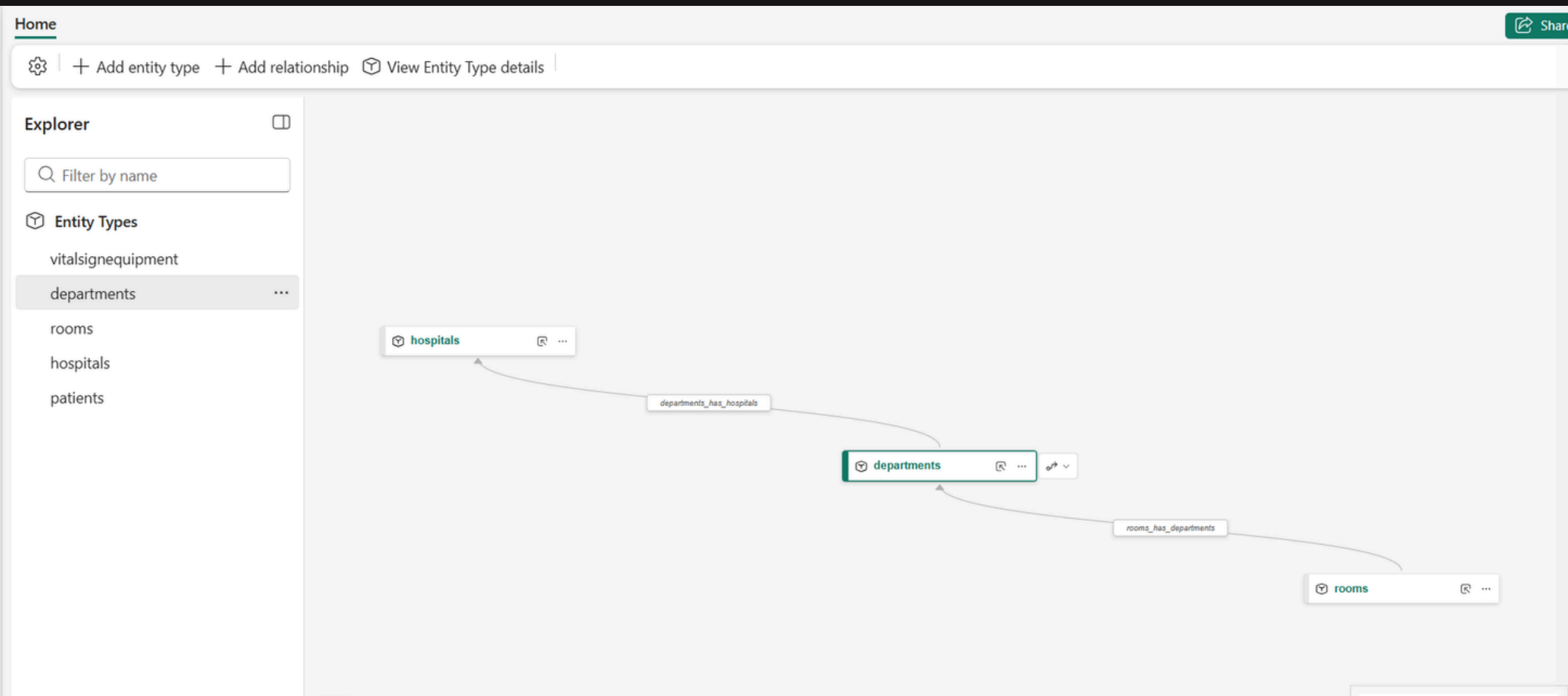
Name *

LamnaHealthcareOntology

Location *

ontology_semantic_model

Create Cancel





DP-600 : Implement analytics solutions using Microsoft Fabric

► Review and add entity type keys

Home > hospitals

Configure Instances Overview

Properties ⓘ Manage property bindings ▾ Choose display name property ▾

Entity type key HospitalId 🔑 Search by property name None ▾

Property name ↑	Type	Data source
City	String	 hospitals
HospitalId 🔑	Integer	 hospitals
HospitalName	String	 hospitals
State	String	 hospitals

Home > departments

Configure Instances Overview

Properties ⓘ Manage property bindings ▾ Choose display name property ▾

Entity type key DepartmentId 🔑 Search by property name None ▾

Property name ↑	Type	Data source
DepartmentId 🔑	Integer	 departments
DepartmentName	String	 departments
Floor	Integer	 departments
HospitalId	Integer	 departments



DP-600 : Implement analytics solutions using Microsoft Fabric

➤ Review and add entity type keys

Home > rooms

Configure Instances Overview

Properties ⓘ Manage property bindings ▾ Choose display name property ▾

Entity type key RoomId None ▾

Property name ↑	Type	Data source
DepartmentId	Integer	rooms
RoomId	Integer	rooms
RoomNumber	String	rooms
RoomType	String	rooms

Workspaces

Copilot

OneLake catalog

Monitor

Real-Time

Workloads

ontology_se
mantic_m...

Configure Instances Overview

Properties ⓘ Manage property bindings ▾ Choose display name property ▾

Entity type key PatientId None ▾

Property name ↑	Type	Data source
AdmissionDate	DateTime	patients
CurrentRoomId	Integer	patients
DateOfBirth	DateTime	patients
FirstName	String	patients
LastName	String	patients



DP-600 : Implement analytics solutions using Microsoft Fabric

▶ Verify and configure relationship types

Home > departments_has_hospitals

departments_has_hospitals
departments > hospitals

Relationship

Origin entity type *
departments

Entity type keys
123 DepartmentId

Mapping table
departments

Relationship name
patients_has_rooms

Target entity type *
rooms

Entity type keys
123 RoomId

Mapping table
rooms

Switch direction

Home > rooms_has_patients

rooms_has_patients
rooms > patients

Relationship

Origin entity type *
patients

Entity type keys
123 PatientId

Mapping table
patients

Relationship name
patients_has_rooms

Mapping table
patients

Matched patients: PatientId *
123 PatientId

Matched rooms: RoomId *
123 CurrentRoomId

Target entity type *
rooms

Entity type keys
123 RoomId

Mapping table
rooms

Switch direction



DP-600 : Implement analytics solutions using Microsoft Fabric

➤ Verify and configure relationship types

Home > vitalsignequipment_has_patients

vitalsignequipment_has_patients
vitalsignequipment > patients

Relationship ↔ Switch direction

Origin entity type *

vitalsignequipment

Entity type keys

ABC EquipmentId

Mapping table

vitalsignequipment

→

Relationship name

vitalsignequipment_has_patients

Mapping table ⓘ

vitalsignequipment

— vitalsignequipment

Matched vitalsignequipment: EquipmentId ⓘ *

ABC EquipmentId

— patients

Matched patients: PatientId ⓘ *

123 PatientId

→

Target entity type *

patients

Entity type keys

123 PatientId

Mapping table

patients



DP-600 : Implement analytics solutions using Microsoft Fabric

➤ Add time-series binding to VitalSignEquipment

Home > vitalsignequipment > Bind data to properties

vitalsignequipment

Entity type key

EquipmentId Edit

Binding selection

☐ vitalsignequipment
ontology_semantic_model > Lamn...

☒ VitalSignsReadings
ontology_semantic_model > Lamn...

Add data binding

Entity type key mapping

Choose source column data to map it to property name(s) used in the entity type key

Source column	Property name
EquipmentId	EquipmentId

Timeseries data ⓘ

One or more columns have been identified as date/time data. Select a column that represents an event or measurement timestamp identifying these properties as timeseries.

Timestamp column ⓘ *

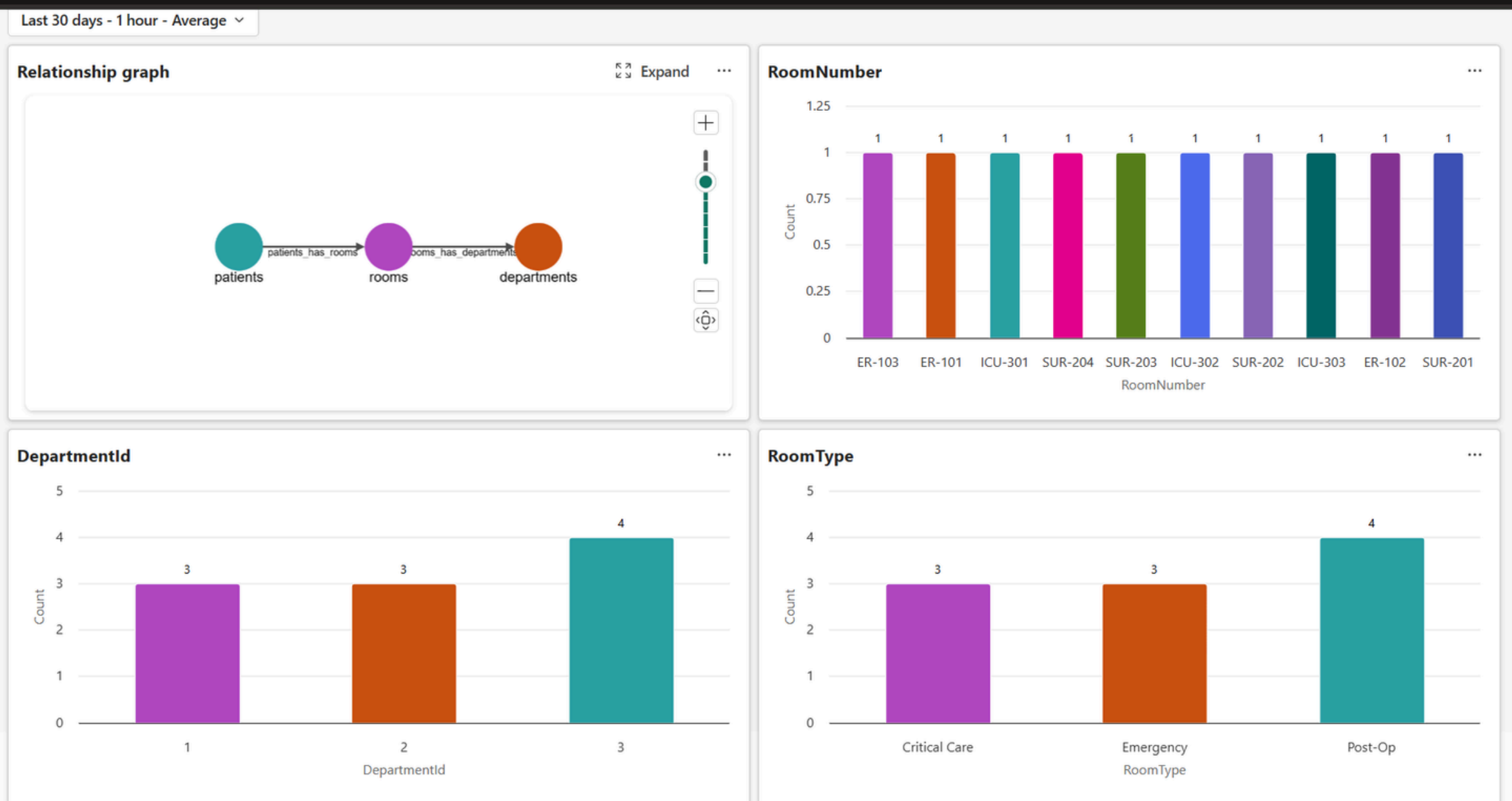
Timestamp

Save Cancel



DP-600 : Implement analytics solutions using Microsoft Fabric

➤ Preview the ontology





DP-600 : Implement analytics solutions using Microsoft Fabric



Reference

